



NC STATE
UNIVERSITY

NRP 
Nuclear Reactor Program

NE235 Nuclear Reactor Operations Module #11 Neutron Flux and Power Density, & Heat Balance and Power Calibration

**Dept. of Nuclear Engineering
North Carolina State University**

NE235 Nuclear Reactor Operations Training

Topics:

- A. Neutron Flux Distribution
- B. Power Density & Peaking Factors
- C. Departure from Nucleate Boiling (DNB) and Safety Limits
- D. Heat Balance & Power Calibration

Laboratory Sessions:

- Lab#7 – Axial Flux Profile - THURSDAY
- Lab#8 – Heat Balance - TUESDAY

Homework #11 – Due 11/29

Neutron Flux & Power Density

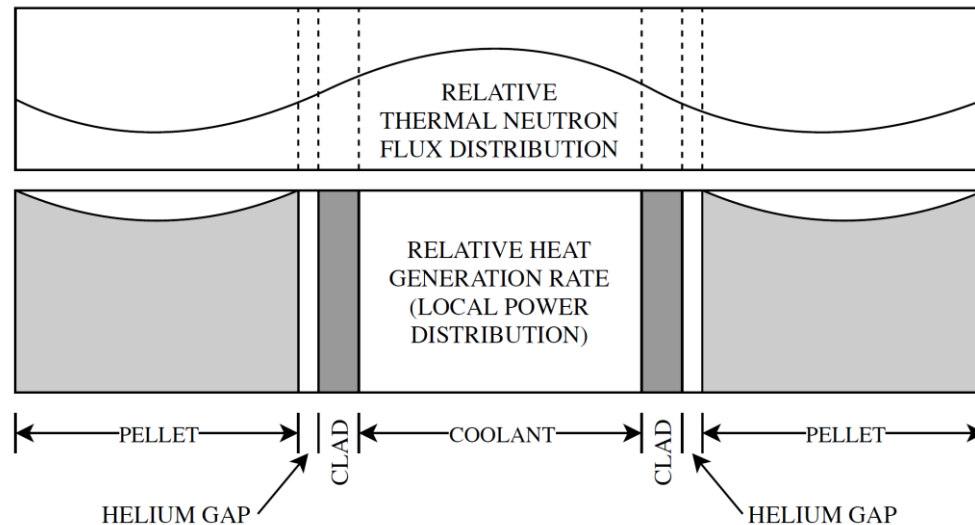
Flux (Φ) and Power Density are variable throughout the core

- Variable across the fuel pellet as the flux dips in the center due to self shielding of thermal neutrons
- Variable radially across the fuel assemblies (horizontally)
- Variable vertically along the fuel channel

Variability in the radial and vertical flux is a function of many factors including:

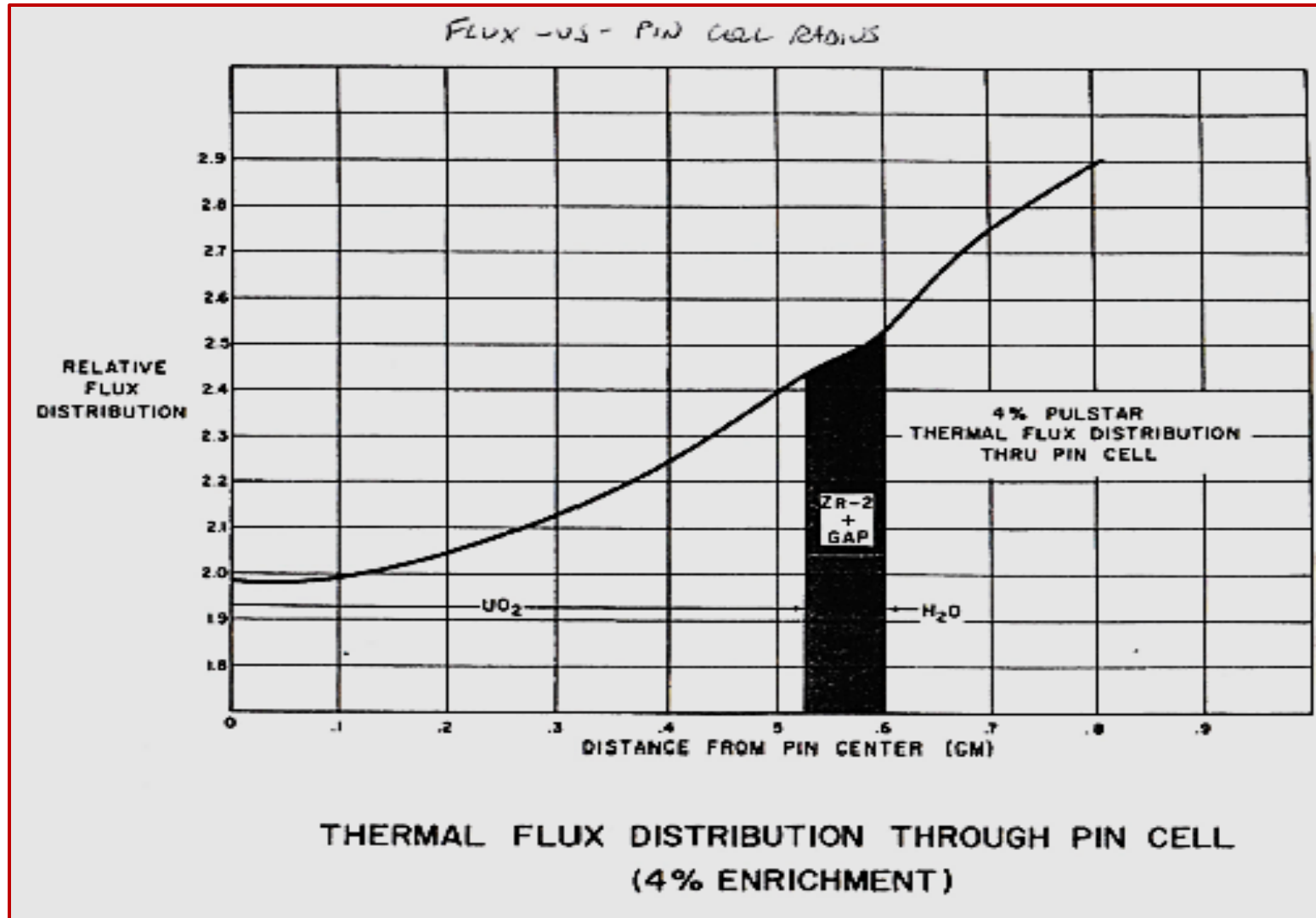
- Power history
- Moderator temperature
- Geometry & characteristics of reflector materials
- Enrichment and Burnup of ^{235}U
- Critical rod height
- Distribution of Fission Product Poisons: ^{135}Xe , ^{149}Sm

Neutron Flux Gradients

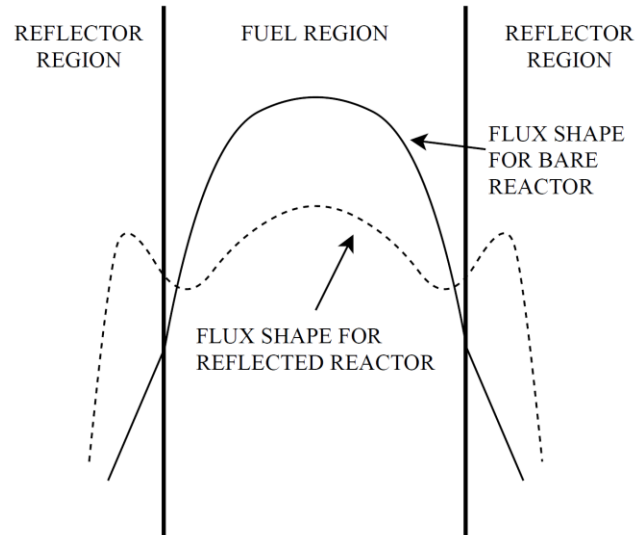


- ❑ Variable across the fuel pellet as the flux dips in the center due to self shielding of thermal neutrons.
- ❑ Peaks in coolant channel in under-moderated core like light water moderated & cooled PULSTAR.
- ❑ Variable radially across the core (horizontally).
- ❑ Thermal peaking in reflector due to fast leakage.

Neutron Flux Gradients: Radially in Fuel Pin

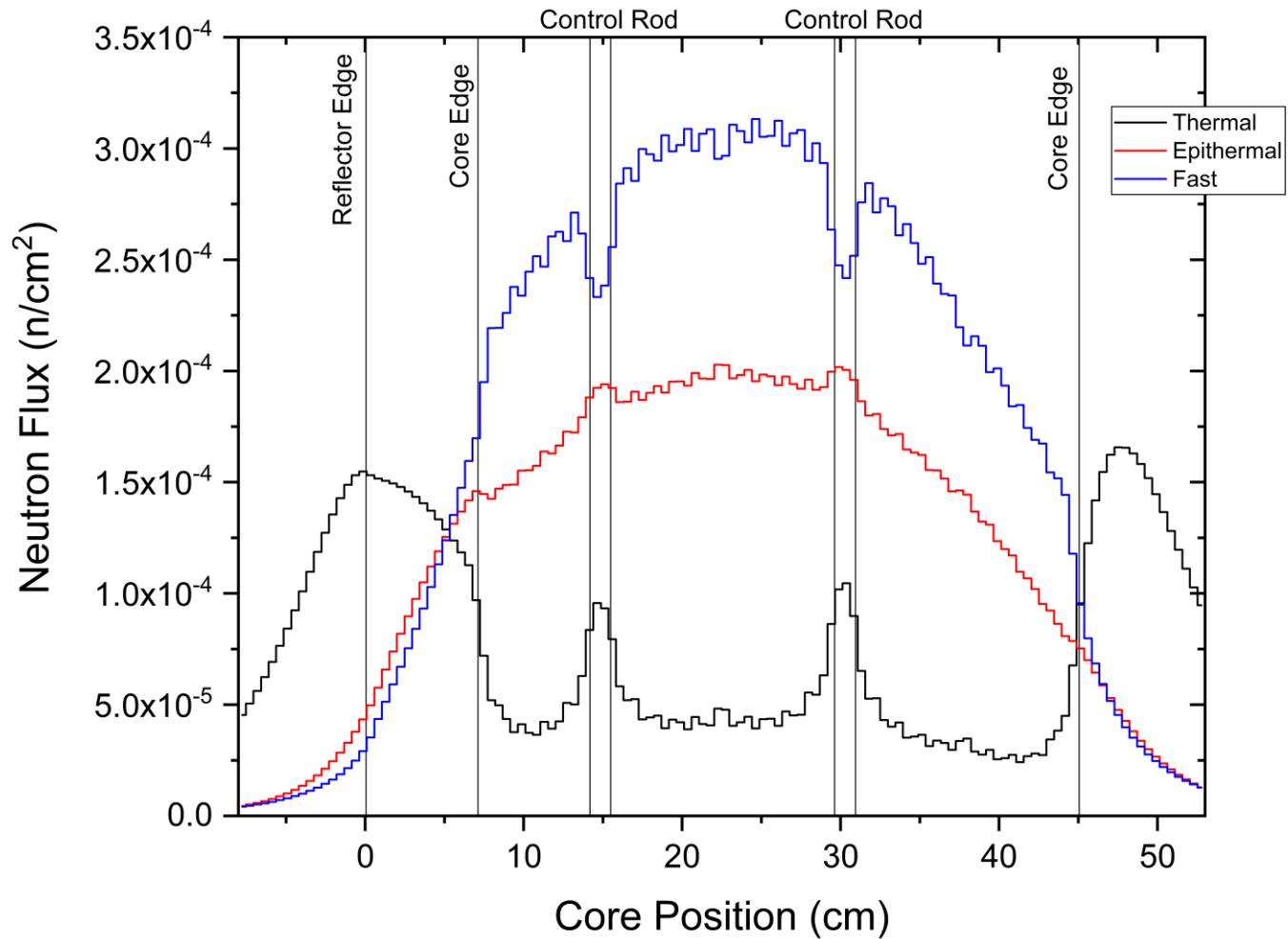


Neutron Flux Gradients

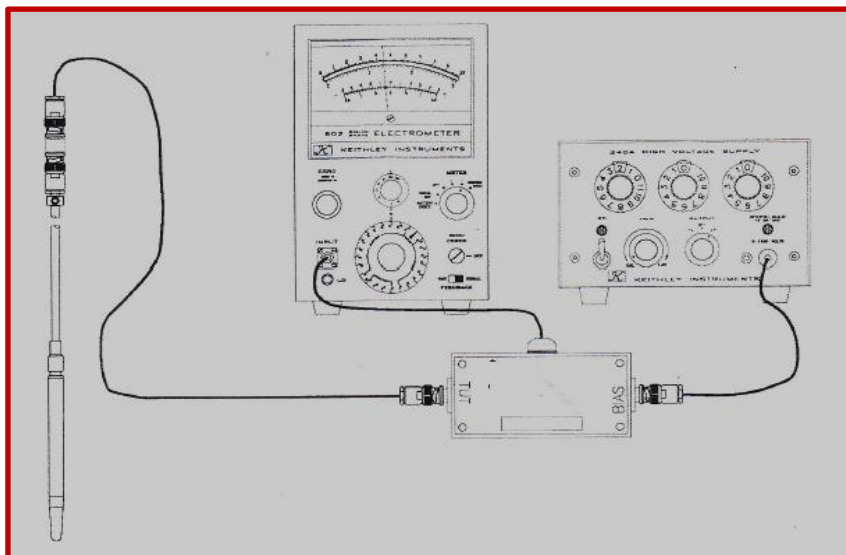


- ☐ Variable across the fuel pellet as the flux dips in the center due to self shielding of thermal neutrons.
- ☐ Peaks in coolant channel in under-moderated core like light water moderated & cooled PULSTAR.
- ☐ Variable radially across the core (horizontally).
- ☐ Thermal peaking in reflector due to fast leakage.

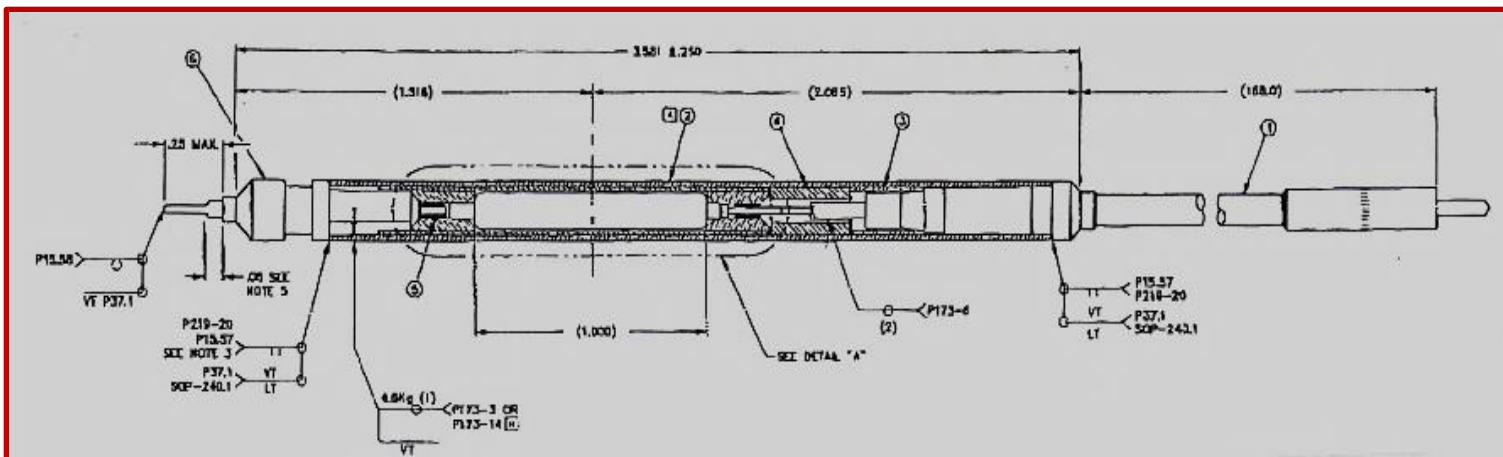
Slide #6 - Neutron Flux Gradients: Radially in Core



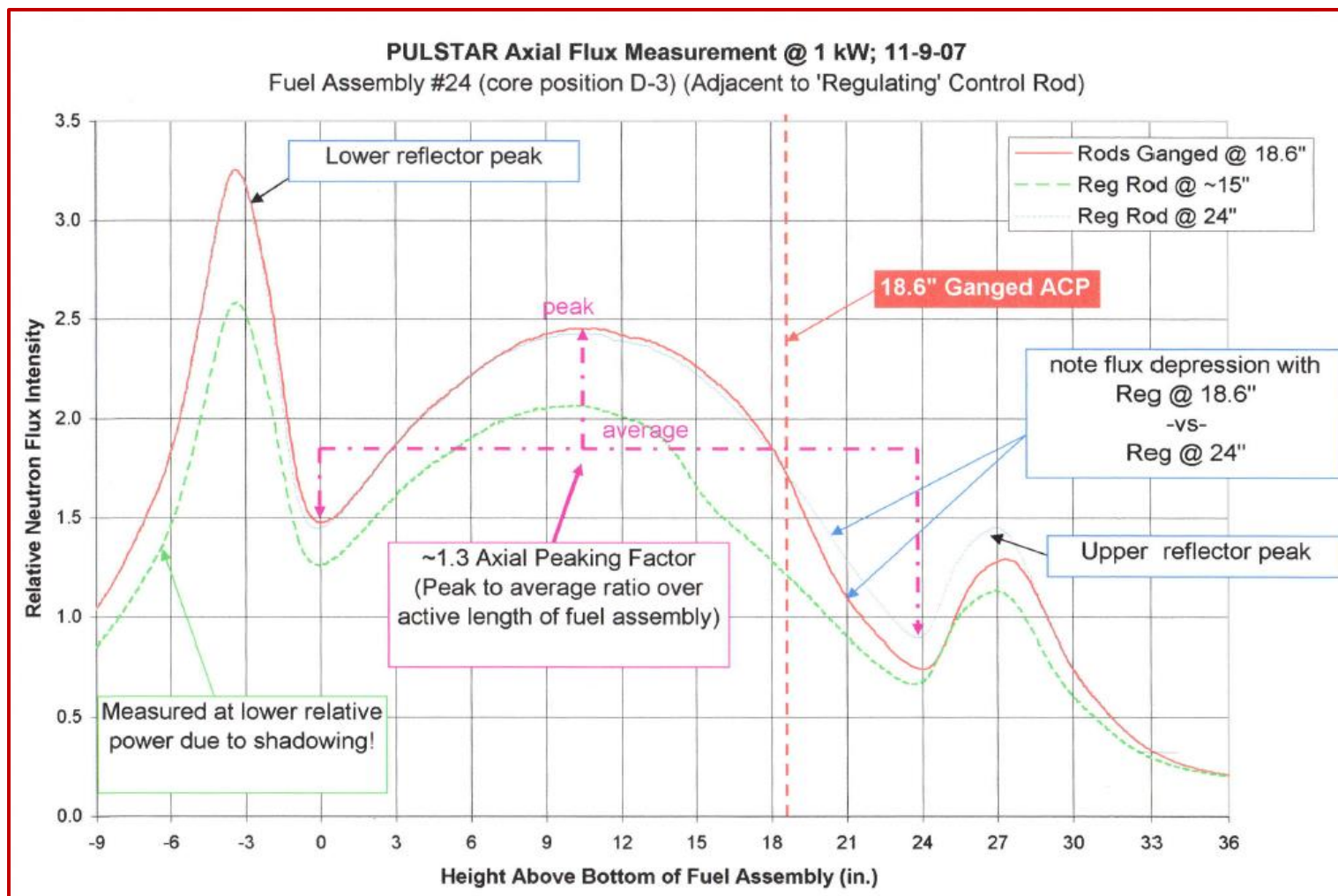
Fuel Assembly Peaking Factor Measurement



- ❑ Miniature In-Core (MIC) fission chambers used to measure flux distribution.
- ❑ Inserted in dry wells inside fuel assemblies.
- ❑ Current generated through ionization of recoiling fission products in detector is proportional to local neutron flux.



Neutron Flux Gradients: Axially (vertically)



Assembly Peaking Factors

FIGURE 16
REFLECTED CORE NO. 4
PEAKING FACTORS

1	2	3	4	5	
23	14	10	11	6	B
0.931	1.340	1.426	1.398	1.321	
20	15	2	3	7	C
1.081	1.594	1.644	1.632	1.527	
21	16	4	5	8	D
1.186	1.704	1.812	1.667	1.472	
26	17	12	13	9	E
1.049	1.555	1.554	1.499	1.363	
24	22	18	19	25	F
0.754	1.190	1.249	1.122	1.036	

Assy #

flux

- ☐ Fuel assembly peaking factor (*axial x radial*) –vs- Pin peaking factor.
- ☐ Measure assembly peaking factors with in-core detectors.
- ☐ Benchmark codes that model pin peaking factors to assure pin peaking is within TS limits within the ‘hot channel’.

Pin Peaking Factor & DNB Limits

PULSTAR Technical Specifications (TS):

T.S. LCO 3.1(f): Maximum (Hot Pin) **Peaking Factor** ≤ 2.92 Peak-to-Average

Maintains local power density (watts/cm³) low enough to **avoid DNB!**

$$\text{Power Density} \left(\frac{\text{Watts}}{\text{cm}^3} \right) = C \left(\frac{\text{Watts} * \text{sec}}{\text{Fission}} \right) * N \sigma_f \left(\frac{1}{\text{cm}} \right) * \Phi_{TH} \left(\frac{n}{\text{cm}^2 \text{sec}} \right)$$

Power density in fuel (watts/cm³) drives heat flux across cladding (watts/cm²), where 'Critical Heat Flux' = onset of '**Departure from Nucleate Boiling**' (DNB).

$$\text{DNB Ratio} \geq 2.0 = \frac{\text{Critical Heat Flux}}{\text{Heat Flux}}$$

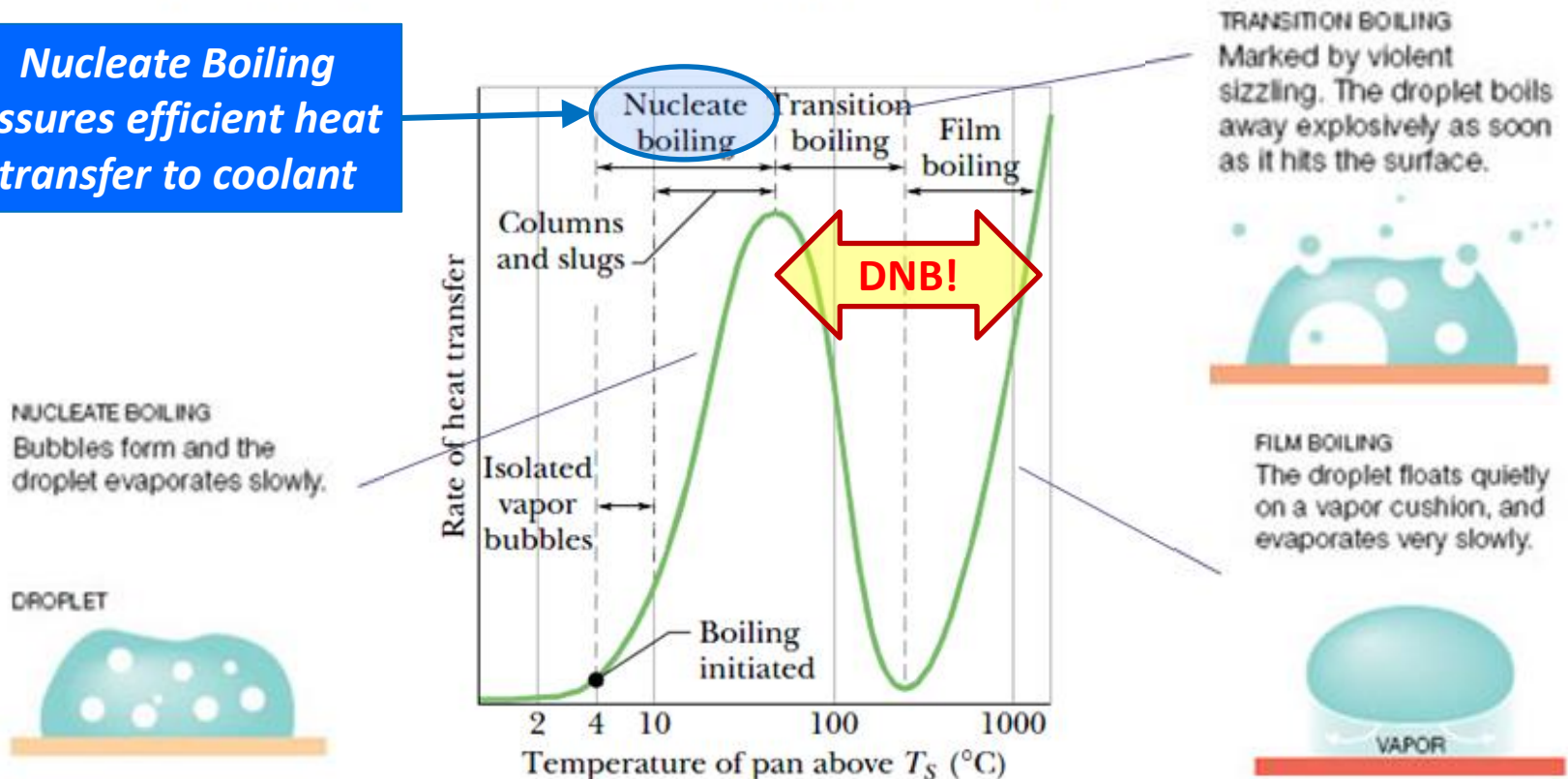
Minimum DNB ≥ 2.0 is a typical limiting safety criterion under all steady state and accident transient conditions.

Heat Transfer & DNB: Nucleate Boiling –vs- Film Boiling

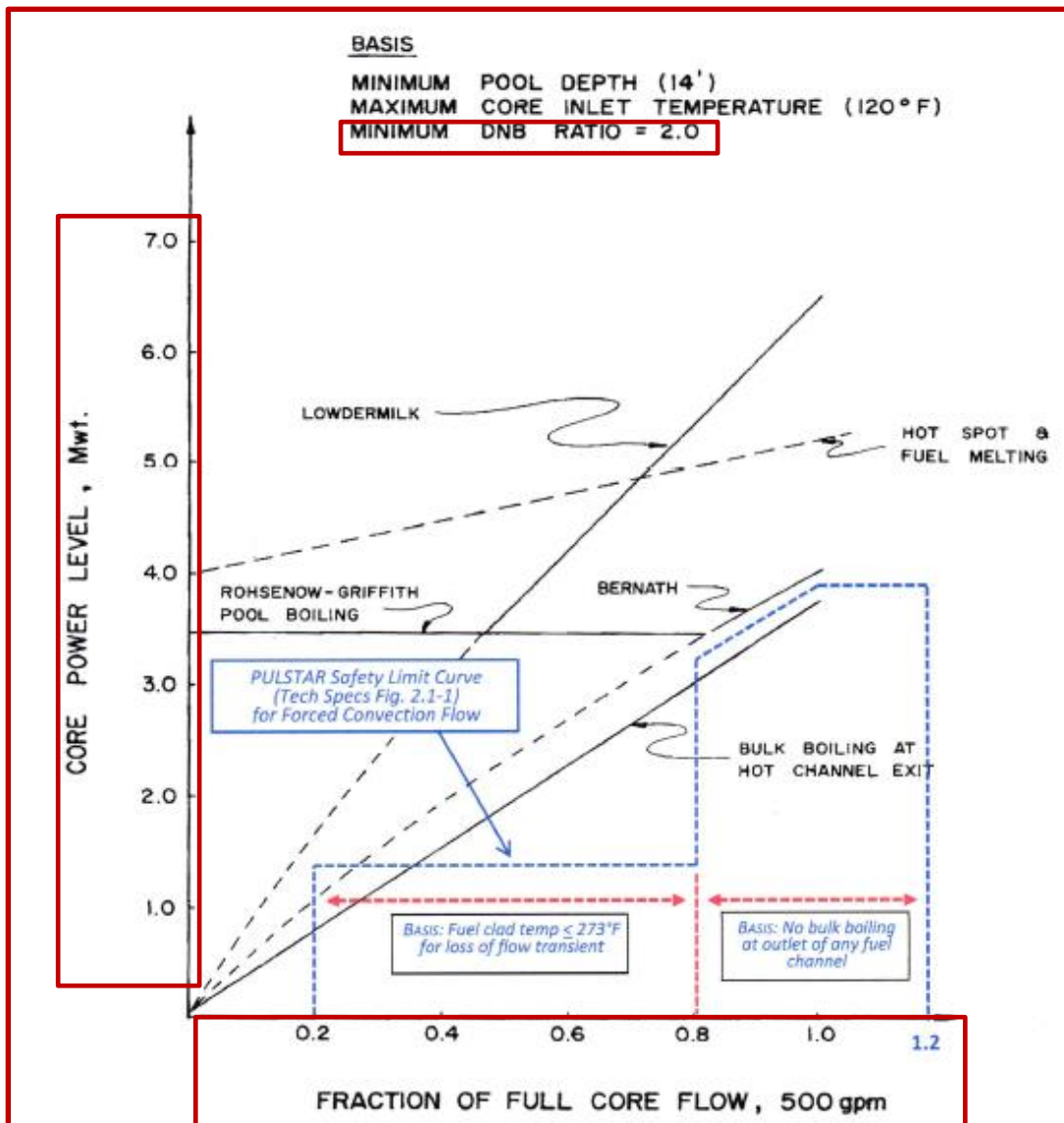
Heat transfer for water (@ 1 atm)

S-shaped graph when heat flux (q'') is compared to temperature.

Nucleate Boiling
assures efficient heat
transfer to coolant



Safety Limit Envelope for PULSTAR



- ❑ **Current SAR Safety Limit:** Core Power –vs- Primary Coolant Flow Curve.
- ❑ Safety Limit ‘envelope’ assures minimum DNB ratio of > 2.0 is always met under steady state and accident transient conditions.
- ❑ *Lower power and higher coolant flow rate required to stay within DNB limits.*
- ❑ **Updated SAR Safety Limit:** Based on Fuel Clad Temp; $< 2,200^{\circ}\text{F}$ under all steady state and accident transient conditions (Reactivity insertion, LOCA, LOFA)

Reactor Heat Balance

Perform Heat Balance to calibrate reactor power and stay within limits:

$$\textit{Heat Gains (In Core)} = \textit{Heat Losses (Around Cooling Loop)}$$

Therefore,

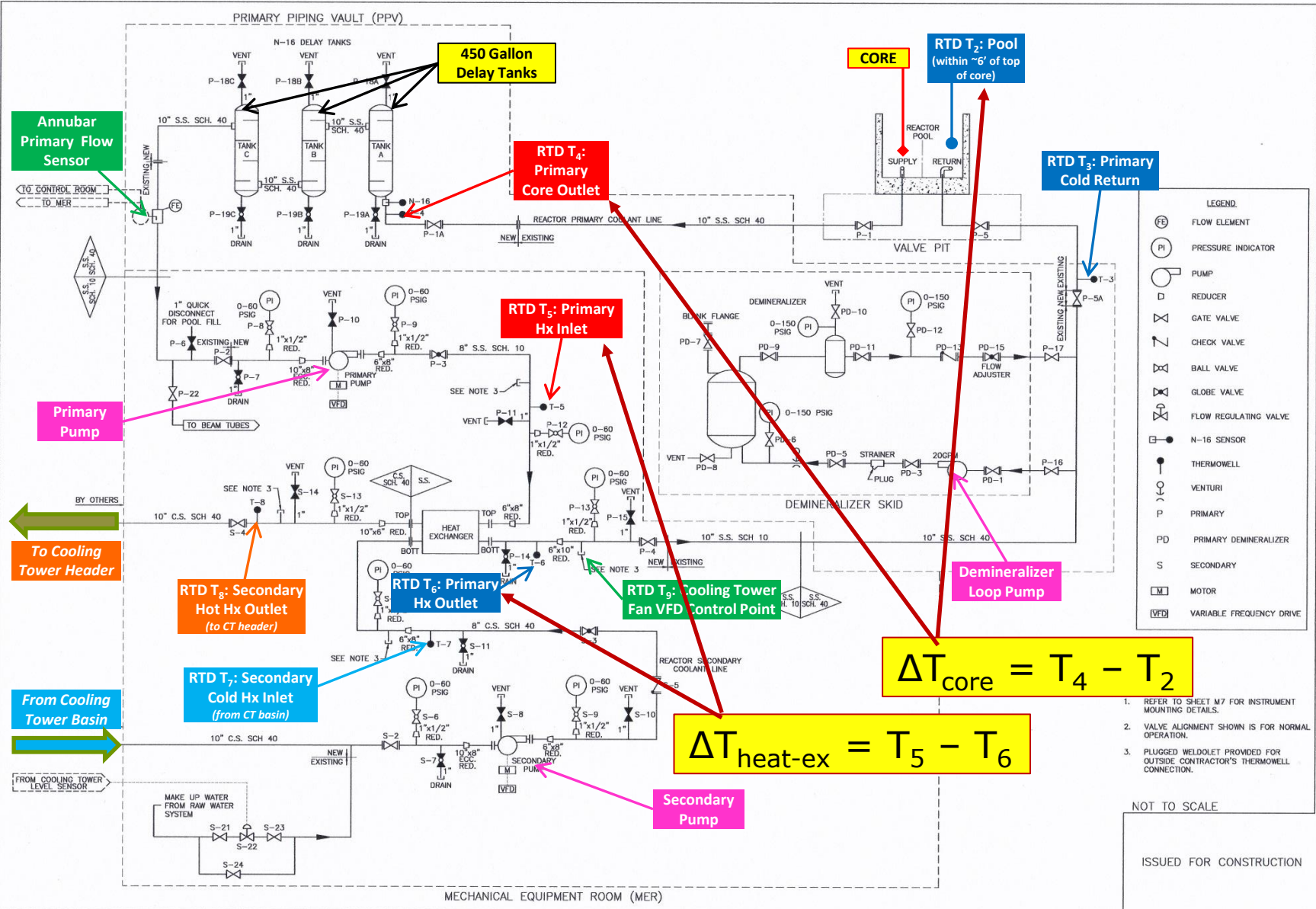
$$\text{Reactor Power} = \text{Heat Exchanger Losses} + \text{Pool Heat-Up} + \text{Ambient Losses}$$

Where:

$$\dot{Q} \left(\frac{\text{Btu}}{\text{hr}} \right) = \underbrace{\dot{m} \left(\frac{\text{lbm}}{\text{hr}} \right)}_{\text{Mass Flow Rate of Primary}} \times \underbrace{C_p \left(\frac{\text{Btu}}{\text{lbm}^\circ\text{F}} \right)}_{\text{Heat Capacity of Water}} \times \underbrace{\Delta T (^\circ\text{F})}_{\text{Across Heat Exchanger}}$$

$$\underbrace{+ m(\text{lbm})}_{\text{Primary Coolant Mass}} \times \underbrace{C_p \left(\frac{\text{Btu}}{\text{lbm}^\circ\text{F}} \right)}_{\text{Temp Increase "Heatup Rate"}} \times \underbrace{\Delta T (^\circ\text{F/hr})}_{\text{Go to } \sim 0 \text{ at Thermal Equilibrium}} + \text{Ambient Losses}$$

PULSTAR - Primary Cooling System Schematic



Heat Balance – Power Calibration

Example: Calculate true reactor power $\dot{Q}(MW)$

Given: ΔT across the heat exchanger = $+10^{\circ}F$

Primary Heatup Rate = $+0.5^{\circ}F/hr$

Primary Volume = $15,650\text{ GAL}$

Primary Flow Rate = 500 GAL/min

Where:

Density of Water is $8.272 \frac{lbm}{GAL}$ at $\sim 108^{\circ}F$ (nominal avg. coolant temp)

Where $1\text{ MW} = 3.413 \times 10^6\text{ Btu/hr}$ (*Btu = British Thermal Unit where 1 Btu of heat raises one pound of water by $1^{\circ}F$*)

Reactor Power Calibration by Heat Balance

$$\dot{Q} = \dot{m}C_p\Delta T + mC_p\Delta T + 0 \quad \text{Eq. from Chpt. 3}$$

$$\dot{Q}\left(\frac{\text{Btu}}{\text{hr}}\right) = \left(500 \frac{\text{GAL}}{\text{min}} \times 8.272 \frac{\text{lbm}}{\text{GAL}} \times 60 \frac{\text{min}}{\text{hr}}\right) \times \left(0.998 \frac{\text{Btu}}{\text{lbm}^\circ\text{F}}\right) \times (+10^\circ\text{F})$$

$$+ \left(15650 \text{ GAL} \times 8.272 \frac{\text{lbm}}{\text{GAL}}\right) \times \left(0.998 \frac{\text{Btu}}{\text{lbm}^\circ\text{F}}\right) \times \left(0.5 \frac{^\circ\text{F}}{\text{hr}}\right)$$

$$= 2.48 \times 10^6 \frac{\text{Btu}}{\text{hr}} + 6.46 \times 10^4 \frac{\text{Btu}}{\text{hr}} = 2.54 \times 10^6 \frac{\text{Btu}}{\text{hr}}$$

$$\dot{Q}_{\text{Reactor}} = \frac{1 \text{ MW}}{3.413 \times 10^6 \frac{\text{Btu}}{\text{hr}}} \times 2.54 \times 10^6 \frac{\text{Btu}}{\text{hr}} = \mathbf{0.745 \text{ MW}}$$

Summary

- ❑ The neutron flux distribution varies throughout core volume – within the fuel pin and both radially and axially in the core.
- ❑ The spatial power density (watts/cm³) in the fuel is a function of the flux distribution and drives the **hot pin** and assembly peaking factors.
Reactor TS limits pin peaking factor to <2.92 peak to average.
- ❑ If the local power density gets too high, the Critical Heat Flux (watts/cm²) across the cladding can get too high and lead to a '**Departure from Nucleate Boiling (DNB)**'
- ❑ Safety Limits for peaking factor, reactor power, and primary coolant flow assure that **DNB $\geq$ 2.0**, **so fuel cladding is not damaged!**
- ❑ A heat balance measurement is performed to calibrate the power level of the reactor to assure it stays within safety limits.